VIDEO GAME CREATION

# Lesson 12 - Hearing It Happen

***Semester 1, Week 12 - Sound Effects***

- **Objectives:** Wire sound cues (shoot, explosion, wave clear, life lost, game over) into the exact code paths that already represent those events; recognize why event-driven trigger points don't need a manual replay guard; formalize sound effects that were optional 'Level 3 stretch' ideas in Lessons 2-8 into a required capstone feature.
- **IGDA Tie-In:** Audio - sound design fundamentals, diegetic vs. non-diegetic cues (introductory); Programming - event-driven triggers, code reuse (intermediate)
- **Featured Engine (lecture demo):** GameMaker (GML Visual - second rotation slot, callback to Lesson 5)
- **Materials:** Demo laptop with all 8 engines verified and audio output working (see Teacher's Checklist); Student Handouts, one per engine; students' own Lesson 11 projects (already saved); one new sound asset (sfx_wave.wav) - no other new assets needed this week

## Agenda

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| 0:00-0:05 | Welcome back, Lesson 11 recap, the polish phase begins |
| 0:05-0:15 | The Sound Payoff: Four Sounds, Four Existing Moments |
| 0:15-0:35 | One Shared Sound for Three Outcomes (live demo in GameMaker) |
| 0:35-0:45 | Why None of Today's Sounds Need a Retrigger Guard |
| 0:45-0:55 | Introduce lab exercise, hand out engine-specific handouts |
| 0:55-1:50 | LAB: Wire up all four sounds, test a full round by ear |
| 1:50-2:00 | Exit check / quick share-out - play a round for the class |

## Levels for this lesson's exercise

- Level 1 (Minimal): every expected sound plays at the correct moment at least once - shoot, explosion/hit, wave clear, life lost, game over.
- Level 2 (Complete): every sound fires reliably every time with no missing or duplicate playback, and none interfere with each other on the same frame.
- Level 3 (Refined): add the optional win sound if not already built, add pitch/volume variety, and/or tidy the sound-triggering code.

*See: Lesson 12 Teacher's Script, Lesson 12 Prep Checklist, and the per-engine Student Handouts. This is a lighter lesson than most - mostly wiring existing branches, not new mechanical complexity. Bring headphones or speakers for the lecture demo.*